

An Introduction to coreBuilder and coreConsultant 1.1

coreBuilder/coreConsultant Application Note

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Getting Started with coreBuilder and coreConsultant

In this section all references to coreBuilder can be replaced with coreConsultant as the tools are invoked in the same way and share the same invocation options.

Invoking the Tools

To invoke coreBuilder in GUI mode, enter the following command at the UNIX prompt:

```
% $SYNOPSYS/sparcOS5/dware/bin/coreBuilder
```

To invoke coreBuilder in command line mode, enter the following command at the UNIX prompt:

```
% $SYNOPSYS/sparcOS5/dware/bin/coreBuilder -shell
```

coreBuilder and coreConsultant Command Line Options

The complete syntax for the coreBuilder and coreConsultant commands are:

```
coreBuilder [<Workspace_name> | <KB_name>] [-shell]  
[-x "<command string>"] [-f <cmd_file>]
```

Table 1

Argument	Description
<Workspace_name>	Loads the workspace KB for <Workspace_name> upon startup.
<KB_name>	Loads the specified .kb file (for example, my_plugin.kb) upon startup.
-shell	Selects command line mode instead of GUI mode. For more information about command line mode, refer to Chapter XXX.
-x "<command_string>"	Executes <command_string> upon startup. <command_string> is one or more coreBuilder commands, separated by semicolons (;).
-f <cmd_file>	Executes the commands in a specified command file (<cmd_file>) in batch mode upon startup. <cmd_file> contains one or more coreBuilder commands. See the "Batch Mode" section of Chapter XXX for more information about batch mode.

Examples

To invoke coreBuilder and automatically load the Developer1 workspace, enter the following command:

```
% coreBuilder <path>/Developer1
```

To invoke coreBuilder and load the plugin named MyPlugin.kb in the current directory, enter the following command:

```
% coreBuilder -x "read_kb MyPlugin.kb"
```

To invoke coreBuilder in command line mode and execute the batch command file named buildcoreKit.tcl, enter the following command:

```
% coreBuilder -shell -f buildcoreKit.tcl
```

coreBuilder and coreConsultant Console

The console (Figure 4-1) appears when you invoke coreBuilder or coreConsultant in GUI mode and, until you create a workspace, is your primary interface to the tool. You can perform operations by selecting menu options or by entering commands at the coreBuilder> prompt.

Table 4-1 lists the console menu options and their corresponding command line mode commands. Figure 4-2 defines the toolbar icons.

Figure 1 coreBuilder Console

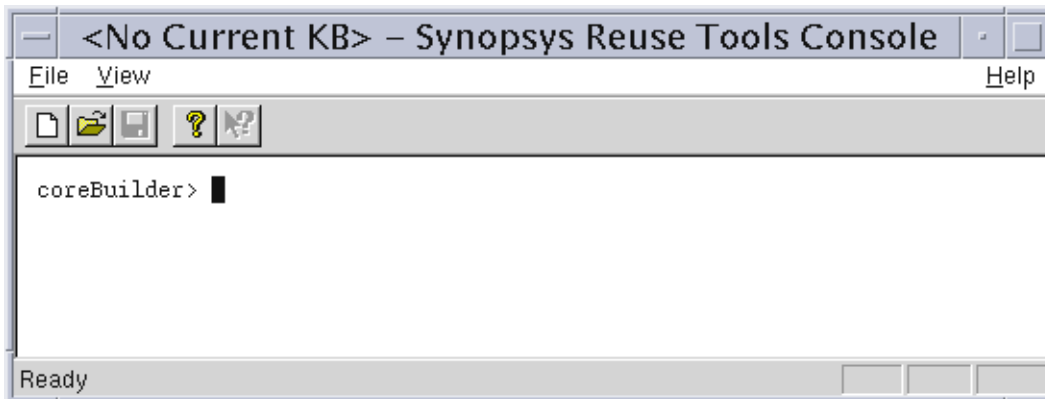


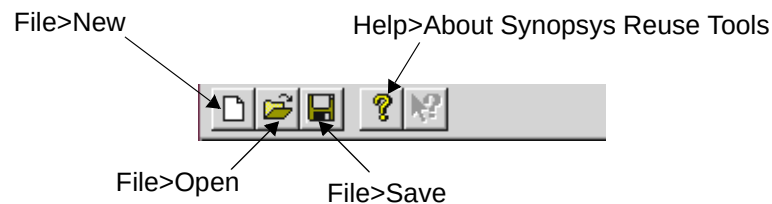
Table 2 Console Menu Commands

Menu Command	Description	Command Line Mode
File>New	Create a new workspace.	create_workspace
File>Open	Open an existing workspace.	open_workspace
File>Save	Save the currently loaded workspace.	save_workspace
File>Save As	Save the currently loaded workspace under a new workspace name.	save_workspace
File>Preferences	Edit user preferences in \$HOME/.synopsys_rt.setup.	None. Edit setup file directly.

Table 2 Console Menu Commands

Menu Command	Description	Command Line Mode
File>Exit	Exit coreBuilder or coreConsultant.	exit or quit
View>Toolbar	Display the console toolbar.	None.
View>Status Bar	Display the console status bar.	None.
Activity>Show Checklist	Display the Activity List for the currently loaded workspace. (Activity menu is available only a workspace is loaded.)	None.
Activity>Express Lane	Invoke Express Lane mode.	None.
Activity><ActivityName>	Initiate the specified activity.	None.
Help>Application Notes	Display a list of available application notes that shipped with this release	None.
Help>Error Messages	Open error message man page index in HTML browser.	None.
Help>User's Guide	Open this user guide index in HTML browser.	None.
Help>Command Reference	Open online reference man page index in HTML browser.	None.
Help>Tutoria	Open the online tutorial in an HTML browser.	None.
Help>About Synopsys Reuse Tools	Display coreBuilder version and copyright information.	None.

Figure 3 Console Toolbar



Quitting coreBuilder or coreConsultant

To quit the tool in GUI mode, select File>Exit in the console. If you have executed commands that modify the contents of a KB, coreBuilder prompts you to save your workspace before exiting. You can either save the KBs and quit, quit without saving modified KBs, or cancel the quit operation.

To quit the tool in command line mode, type `exit` or `quit` at the `coreBuilder>` prompt and press Return. If you execute `exit` or `quit` in the coreBuilder console and you have executed commands that modify the contents of a KB, coreBuilder prompts you to save your workspace before exiting. In shell mode, coreBuilder does not prompt you to save your workspace. If you want to save your work before quitting, enter the `save_workspace` command before either `quit` or `exit`.

coreBuilder and coreConsultant Workspaces

coreBuilder and coreConsultant use local directories to store and manipulate data. This area is referred to as a workspace. You create a workspace after the invocation of coreBuilder or coreConsultant. This is the area that you will build your coreKit when using coreBuilder.

Users of coreConsultant make a unique configuration of the core contained within the coreKit when integrating the core into their design.

Workspace Directory Structures

coreBuilder and coreConsultant automatically creates the directory structure shown in the following diagrams under the selected root directory.

The following table defines each of the directories created by coreBuilder or coreConsultant

Table 4 .

Directory	Description
batch	The directory to store the .tcl files that coreBuilder and coreConsultant automatically generate when you perform design flow activities through the GUI.
install	The directory where coreBuilder stores your final coreKit file. The coreKit in this directory is the file that you would give to the user of your core.
kb	The directory where coreBuilder and coreConsultant stores .kb files for your workspace
report	The directory where coreBuilder and coreConsultant stores reports that are generated while running.
scratch	The directory where coreBuilder and coreConsultant store temporary files.
src	The directory where coreConsultant writes out the core HDL files after the user completes the configuration activity.
syn	This directory is created by coreConsultant to perform synthesis of the core. The user should look in this directory to see synthesis files, i.e. the Synopsys Design Compiler log file for their synthesis run.
test	The default directory where coreBuilder creates a workspace to run coreConsultant when using the “Create Ingetration Workspace” activity. The user can change the name and location of this directory and the configuration name when using this activity. Under the test directory, the user will find a kit_install directory and the configuration directory. The kit_install directory is the place where coreBuilder does a temporary installation of the coreKit for testing purposes. The config1 directory contains the configuration of the coreKit that coreConsultant is using during this activity.

Figure 5 *coreBuilder Workspace Directory Structure*

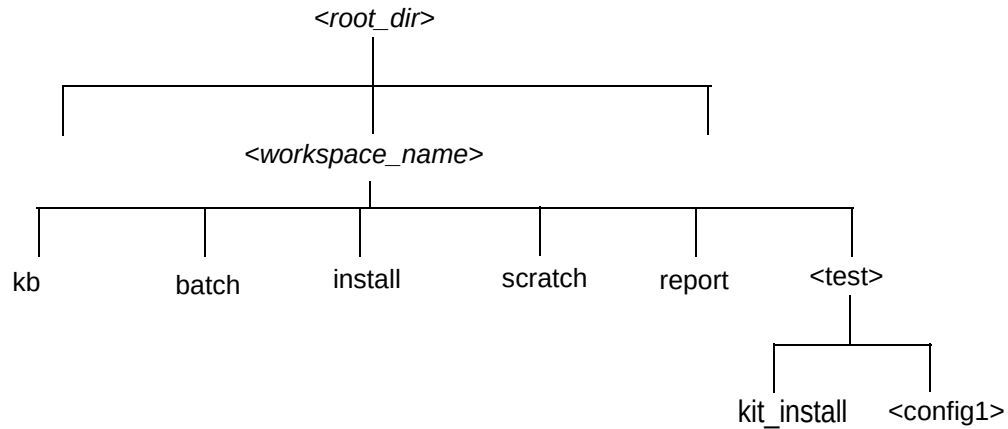
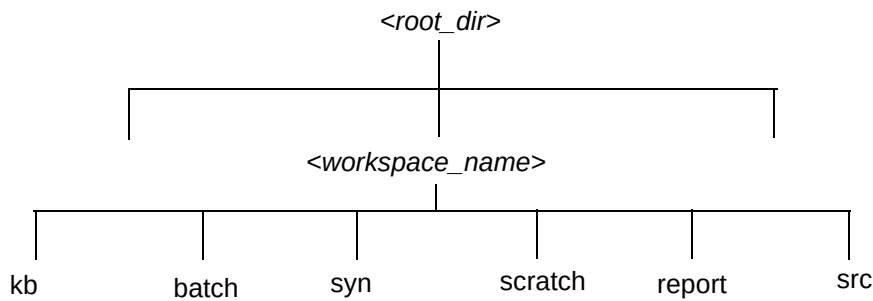


Figure 6 *coreConsultant Workspace Directory Structure*



Basic Workspace Operations

You can save and close your workspace at any time, then re-open your workspace to continue your work. The following sections describe how to create, save, close, open, and move a coreBuilder workspace.

Creating a new Workspace

You can create a workspace interactively in GUI mode or by specifying command line options in command line mode. To create a new workspace in GUI, select File>New from the coreBuilder console menu bar. To create a workspace in command line mode, enter the following command:

```
create_workspace [-gui] [-name <name>] [-root <root dir>]  
[-show_checklist]
```

Table 7

Argument	Description
-gui	Display Create Workspace dialog to create workspace as described for GUI mode. The -gui option works only when you execute create_workspace in the coreBuilder console.
-name <name>	Specifies the workspace name and workspace .kb filename.
-root <root dir>	Specifies the workspace root directory.
-show_checklist	Display Activity List after creating the workspace, works only when you execute create_workspace in the coreBuilder console.

Saving a Workspace

coreBuilder automatically saves all modified KBs, including the workspace KB, each time you complete a coreBuilder activity that modifies the contents of a KB. Therefore, in most cases, you do not need to explicitly save your workspace. However, you may want to save your workspace for either of the following reasons:

- You have executed discrete commands like `set_port_attribute` and have not completed the associated activities (for example, Specify Port Intent), but you want to save your work.

To save the currently loaded workspace in GUI mode, select File>Save in the coreBuilder console.

To save the currently loaded workspace in command line mode, enter the following command:

```
save_workspace
```

- You want to create a duplicate version of your workspace under a new name.

To save the currently loaded workspace under a new name in GUI mode, select File>Save As in the coreBuilder console, then select a new directory (workspace name) for the new workspace.

To save the currently loaded workspace under a new name in command line mode, enter the following command:

```
save_workspace -as <new_workspace_name>
```

Closing a Workspace

To close your workspace in GUI mode, select File>Close in the coreBuilder console. If you have executed commands that modify the contents of a KB, coreBuilder prompts you to save your workspace before exiting.

To close your workspace in command line mode, use the following command:

```
close_workspace [-save]
```

Table 8

Argument	Description
-save	Save workspace before closing.

Opening a Workspace

To open an existing workspace when you invoke `coreBuilder` or `coreConsultant`, specify the workspace name on the command line. If the workspace is not located in the current directory, specify the workspace path name. For example, to open the workspace Builder in the MyCore directory, enter the following command at the UNIX prompt (%):

```
% coreBuilder MyCore/Builder
```

To open a workspace when `coreBuilder` is running in GUI mode, select File>Open in the `coreBuilder` console.

To open a workspace in command line mode, enter the following command:

```
open_workspace <workspace_name>
```

For example, to open the workspace Builder in the MyCore directory, enter the following command:

```
coreBuiler> open_workspace MyCore/Builder
```

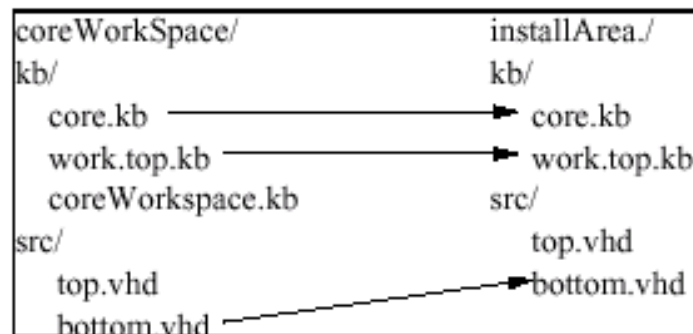
Moving a Workspace

In `coreConsultant` workspaces, files that are common to all configurations are represented as symbolic links back to the installation area to save space. If this installation area is moved, then

the workspace can no longer be opened. To move a coreConsultant workspace to a different location, you can use the UNIX mv command to relocate the workspace directory, but then you have to update the workspace to compensate for the broken links, otherwise the workspace can no longer be opened.

The figure below is an example of a typical workspace with links back into the installation area for the core. Files that are specific to a given configuration (i.e. top.vhd) are in the workspace. If this installation area is moved, then the workspace can no longer be opened since the links will be broken.

Figure 9



To fix the links the user can use the new update_workspace_links which can point the links to a new area, or it can remove the links by copying the linked files into the workspace.

```
update_workspace_links [-copy] [-moved_install <dir>]
<workspace>
```

Table 10

Argument	Description
-copy	Copy all files linked to the installation. This option is mutually exclusive of -moved_install

Argument	Description
-moved_install dir	New location of the installation. This option is mutually exclusive of -copy
workspace	workspace to change